LEIC-T 2025/2026
Aprendizagem - Machine Learning
Homework 2
Deadline 8/10/2025 23:00
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I) Bayesian Classifier (8 pts)

Given a data set describing a sample

x_1	x_2	Class
-2	1	A
-1	2	A
1	3	A
2	4	A
-1	0.4	B
2	1.1	B
3	1.5	B
4	1.8	B

And the query vector $x = (1,2)^T$

a) (3pts) Compute the most probable class for the query vector, under the Naive Bayes assumption, using 1-dimensional Gaussians to model the likelihoods. (Hint, the likelihood is described of each class is described by two Gaussians (Normal Distributions, each distribution is defined by a mean value and standard deviation.)
You can (should?) use your computer with Python, NumPy, MATLAB, Octave, Mathematica, etc. (whatever tool/language you like). Please indicate your results step by step.

Solution:

Gaussian distribution or normal is defined by *the relative probability*

$$p(x|\mu,\sigma^2) = \mathcal{N}(x|\mu,\sigma^2) = \frac{1}{\sqrt{2\cdot\pi}\cdot\sigma}\cdot\exp\left(-\frac{1}{2\cdot\sigma^2}\cdot(x-\mu)^2\right)$$

We have two classes and two independent variables. Each likelihood of class is specified by two Normal distributions. (Naïve assumption)

$\mu_{A1} = (-2-1+1+2)/4 = 0$

$\mu_{A2} = 1+2+3+4)/4 = 2.5$

$\sigma_{A1} = (((-2-0)^2+(-1-0)^2+(1-0)^2+(2-0)^2)/3)^{1/2} = \sqrt{\frac{10}{3}}$

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$\sigma_{A2}=(((1-2.5)^2+(2-2.5)^2+(3-2.5)^2+(4-2.5)^2)/3)^{1/2}=1.290995$

$\mu_{B1}=(-1+2+3+4)/4=2$

$\mu_{B2}=(0.4+1.1+1.5+1.8)/4=1.2$

$\sigma_{B1}=(((-1-2)^2+(2-2)^2+(3-2)^2+(4-2)^2)/3)^{1/2}=\sqrt{\frac{14}{3}}$

$\sigma_{B2}(((0.4-1.2)\text{^}2+(1.1-1.2)\text{^}2+(1.5-1.2)\text{^}2+(1.8-1.2)\text{^}2)/3)\text{^}(1/2)=0.60553$

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p(A)=0.5, p(B)=0.5

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p(A, xquery) = p(xquery,1|A1) · p(xquery,2|A2) · p(A)= 0.18807* 0.28669*0.5=

= 0.026959

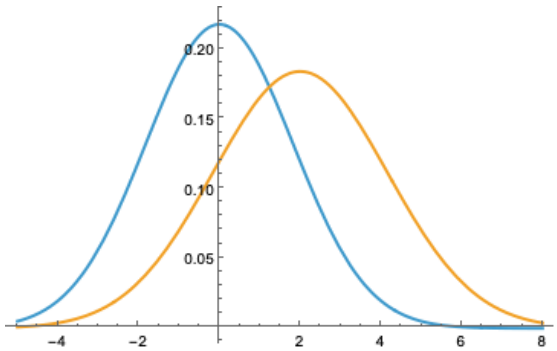
p(B, xquery) = p(xquery,1|B1) · p(xquery,2|B2) · p(B) = 0.16591*0.27526*0.5=

= 0.022835

p(A|xquery) = p(A, xquery)/ (p(A,xquery) +p(B,xquery))= **0. 54141**

p(B|xquery) = p(A, xquery)/ (p(A,xquery) +p(B,xquery))= 0.45859

The vector *x =(1,2)^T* belongs to the class A.



First dimension (x1) of Normal distribution for class A and B

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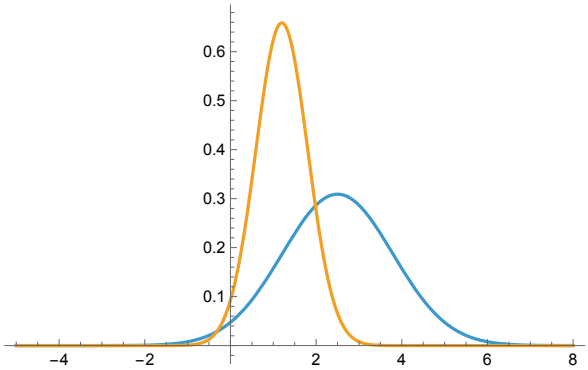

Second dimension (x2) of Normal distribution for class A and B

b) (3 pts) Compute the most probable class for the query vector assuming that the likelihoods are 2-dimensional Gaussians. Are the results from 1 a) and 1 b) the same? Why are the same or not (one sentence, no mathematical proof required)

Solution:

Over D dimensional space

$$p(\mathbf{x}|\boldsymbol{\mu},\Sigma) = \mathcal{N}(\mathbf{x}|\boldsymbol{\mu},\Sigma) = \frac{1}{(2\cdot\pi)^{D/2}}\cdot\frac{1}{|\Sigma|^{1/2}}\cdot\exp\left(-\frac{1}{2}\cdot(\mathbf{x}-\boldsymbol{\mu})^T\Sigma^{-1}\cdot(\mathbf{x}-\boldsymbol{\mu})\right)$$

where

- $\boldsymbol{\mu}$ is the D dimensional mean vector
- Σ is a $D \times D$ covariance matrix
- $|\Sigma|$ is the determinant of Σ

$\mu_A = (0, 0.5)^T$

$$ca11 = ((-2-0)^2 + (-1-0)^2 + (1-0)^2 + (2-0)^2)/3 = \frac{10}{3}$$

$$ca22 = ((1-2.5)^2 + (2-2.5)^2 + (3-2.5)^2 + (4-2.5)^2)/3 = 1.66667$$

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$ca12 =$
$= ((-2-0)*(1-2.5)+(-1-0)*(2-2.5)+(1-0)*(3-2.5)+(2-0)*(4-2.5))/3$
$= 2.33333$

$$\Sigma_A = \begin{pmatrix} \frac{10}{3} & 2.33333 \\ 2.33333 & 1.66667 \end{pmatrix}$$

$$\Sigma^{-1}{}_A = \begin{pmatrix} 15. & -21. \\ -21. & 30. \end{pmatrix}$$

Det(A)=0.111111

$\mu_B = (2, 1.2)^T$

$cb11 = (-1-0.5)\text{^}2 + (0-0.5)\text{^}2 + (1-0.5)\text{^}2 + (2-0.5)\text{^}2)/3 = 1.66667$

$cb22 = ((0.4-1.2)\text{^}2 + (1.1-1.2)\text{^}2 + (1.5-1.2)\text{^}2 + (1.8-1.2)\text{^}2)/3 = 0.366667$

$cb12 =$

$((-1-0.5)*(0.4-1.2)+(0-0.5)*(1.1-1.2)+(1-0.5)*(1.5-1.2)+(2-0.5)*(1.8-1.2))/3$
$=0.766667$

$$\Sigma_B = \begin{pmatrix} 4.66667 & 1.3 \\ 1.3 & 0.366667 \end{pmatrix}$$

$$\Sigma^{-1}{}_B = \begin{pmatrix} 17.3684 & -61.5789 \\ -61.5789 & 221.053 \end{pmatrix}$$

Det(B)=0.021111

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p(A, xquery) = p(xquery,1|A) · p(A) = 0.00002783365166594589*0.5= 0.0000139

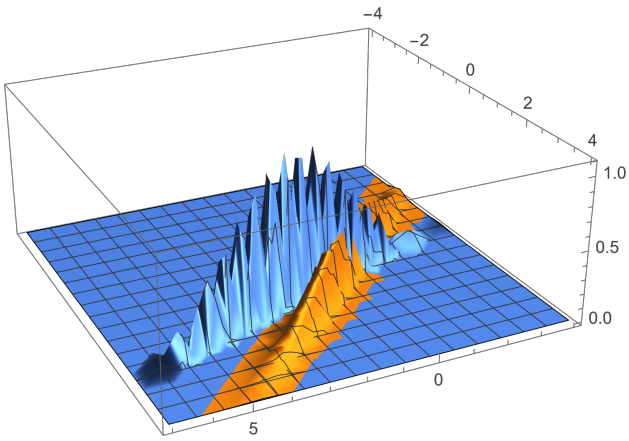
p(B,xquery)=p(xquery,1|B)·p(B) = 1.4214×10^{-56}*0.5= 7.107×10^{-57}

p(A|xquery) = p(A, xquery)/ (p(A,xquery) +p(B,xquery))≈ **1**

p(B|xquery) = p(B, xquery)/ (p(A,xquery) +p(B,xquery))≈ 5.11×10^{-52}

The vector *x =(1,2)*^{T} belongs to the class A.

The result is different, since the dimension x_1 and x_2 are correlated.



c) (1 pts)

x_3	Class
0	A
1	A
1	A
0	A
1	B
1	B
0	B
1	B

And the query vector *x3 =True=1*

Compute the most probable class, with x3 being a categorial class 1=True, 0=False.

Solution:

p(A|1)=card(A.1)/card(1)=2/5=0.4

P(B|1)= card(B.1)/card(1)=3/5=0.6

Most probable class is B

See some additional info if you are confused when you compute d)

P(A|1)= P(1|A)*P(A)/P(1)=P(A,1)/P(1)

P(1|A)= P(A|1)*P(1)/P(A)=P(A,1)/P(A)

d) (1pts) Given a data set describing a sample combining the data set before

x_1	x_2	x_3	Class
-2	1	0	A
-1	2	1	A
1	3	1	A
2	4	0	A
-1	0.4	1	B
2	1.1	1	B
3	1.5	0	B
4	1.8	1	B

x1 and x2 are dependable and x3 is independent of x1 and x2. x3 is a categorial class.
And the query vector $x = (1,2,1)^T$ Compute the most probable class and indicate the estimated relative probability.

Hint,

p(A, xquery) = p((1,2)|A) ·P(1|A) ·p(A)

p(B, xquery) = p((1,2)|B) ·P(1|B) ·p(B)

you have already computed the values in b) and in c)

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$P(1|A)$ =card(A.1)/card(A)=2/4=0.5
$P(1|B)$ =card(B.1)/card(B)=3/4=0.75

Solution:

$p(A, xquery) = p(xquery,1|A) \cdot p(1|A) \cdot p(A) = 0.00002783*2/4*0.5= 0.0000069584$

$p(B, xquery) = p(xquery,1|B) \cdot p(1|B) \cdot p(B) =1.4214 \times 10^{-56}*3/4*0.5=5.33 \times 10^{-57}$

$p(A|xquery) = p(A, xquery)/ (p(A,xquery) +p(B,xquery)) \approx 1$
$p(B|xquery) =1- p(A|xquery) \approx 7.660 \times 10^{-52}$